THE QUESTION

Can I predict heart disease?

Using clinical measurements to identify patterns in 918 patient records.

From patient data to a prediction

A machine learning project by Fatma Çakir

WHY THIS MATTERS

The goal: identify patients at higher risk

The model predicts the dataset’s HeartDisease label, not a medical diagnosis.

INPUT

11 clinical indicators

Age • chest pain • cholesterol

Heart rate • ECG • exercise symptoms

PREDICTION

HeartDisease = 0 or 1

A binary classification task

Useful for exploring screening patterns

THE DATASET

918 records. Two outcome groups.

I used the Kaggle Heart Failure Prediction dataset suggested by my teacher.

918

Patient records

508

Labeled with heart disease

410

Labeled without heart disease

The two groups are reasonably represented, but this is still a limited dataset.

DATA QUALITY

Some measurements were not realistic

I treated these unrealistic zero values as missing measurements.

172

Cholesterol values recorded as zero

1

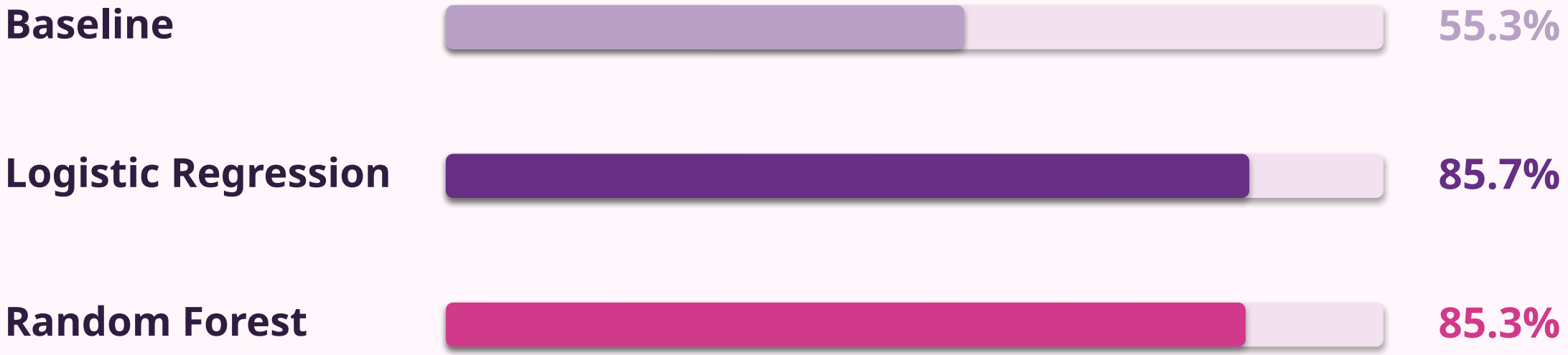
Resting blood pressure value recorded as zero

I imputed missing values and encoded categories inside the training pipeline.

MODEL EXPERIMENT

A simple model was already competitive

Five-fold cross-validation results on training data.



Logistic Regression had slightly higher initial accuracy; Random Forest offered flexibility for tuning.

Could Random Forest do better?

GridSearchCV tested settings such as tree depth and number of trees.

5-fold CV

Repeated validation on training data to compare settings.

0.933

Best cross-validation ROC-AUC for tuned Random Forest.

MAIN FINDING

The tuned model found 94% of positive cases

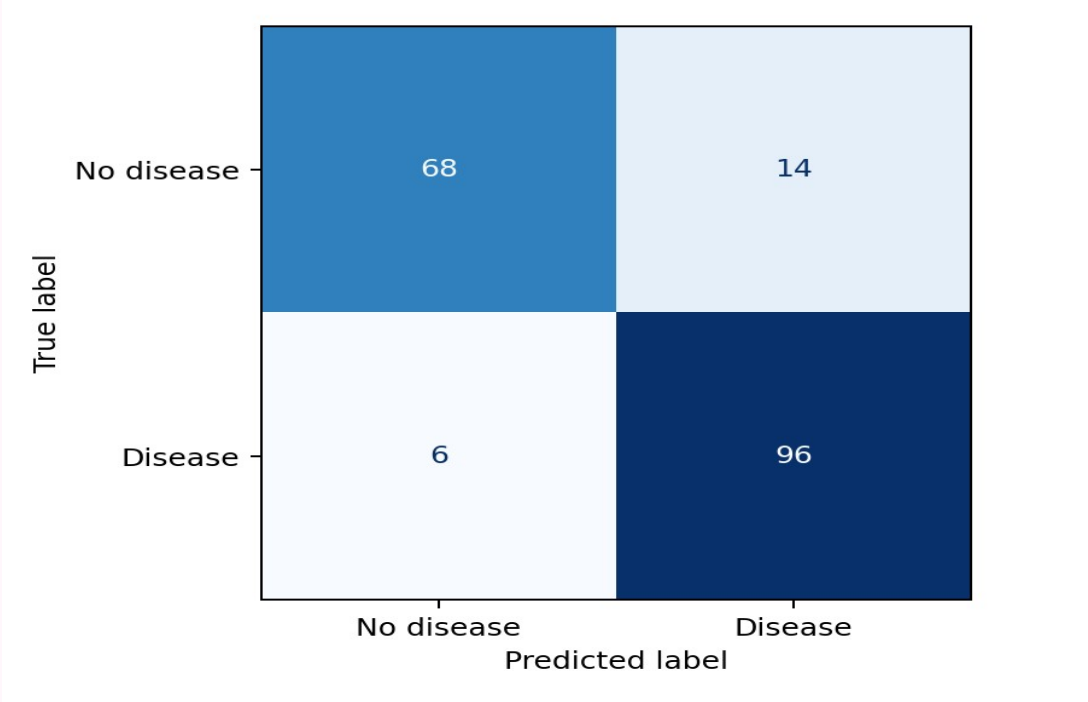
Results on 184 held-out records — data not used for training.

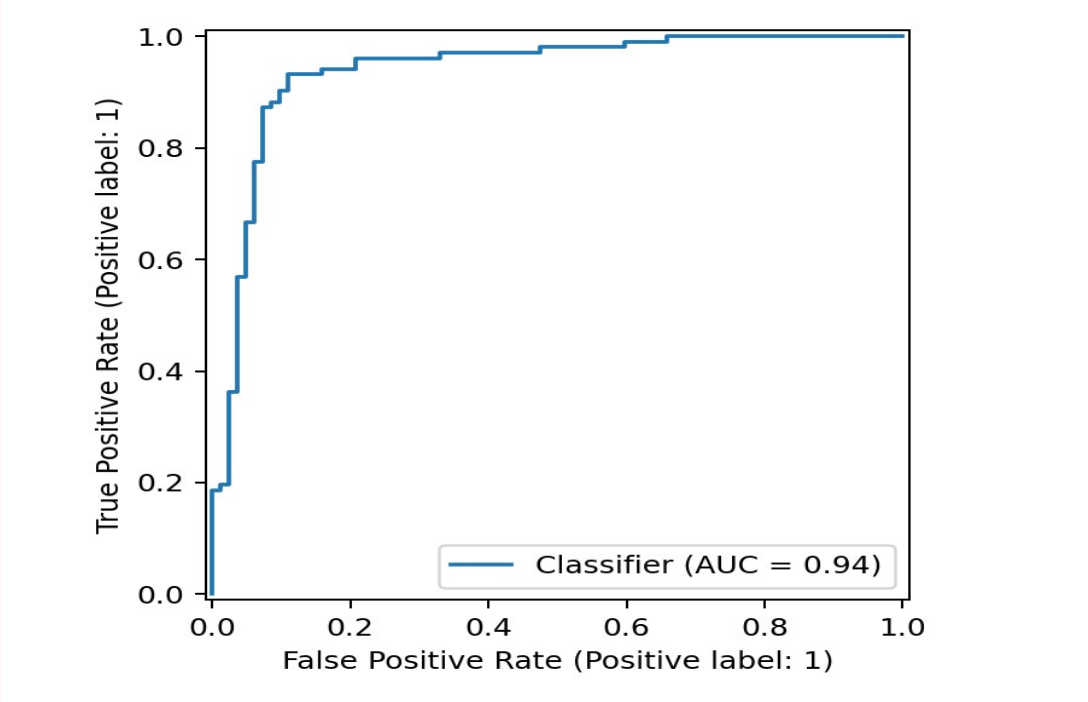
89.1%	94.1%	0.938
Accuracy	Recall	ROC-AUC

Recall matters here: it measures how many truly positive cases the model identifies.

Where did my model make mistakes?

Confusion matrix and ROC curve show what one accuracy number cannot.





Mistakes still occur; evaluating false negatives is essential.

THE MEANING

Promising patterns ≠ clinical readiness

A strong score on one dataset is not enough to support medical decisions.

WHAT I FOUND

Clinical variables contain predictive signal.

Tuning improved the selected model.

WHAT IS STILL NEEDED

External validation

Fairness checks

Calibration and clinician review

THANK YOU

What I take away

Machine learning can reveal useful patterns — but context and evaluation matter.

89.1% accuracy • 94.1% recall

A strong classroom result, with important limitations

Fatma Çakir · Questions?